Cost-Constrained Edge Removal for Epidemic Containment on Social Networks

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Abstract

We study cost-constrained epidemic containment on the Facebook100 social graphs under stochastic SIS dynamics. Per-edge cost is a similarity-based proxy for the social expense of severing a tie; the optimization is to **minimize the total cost of removed edges subject to keeping steady-state SIS prevalence below a target**. We compare four edge-removal policies (random, edge betweenness over cost, a Louvain-induced distance threshold, and a graph neural network whose per-edge cost is learned end-to-end via self-supervised link prediction) across five campuses and four basic reproduction numbers. The headline finding is a regime split. Near the spectral threshold, the GNN cuts steady-state prevalence by 4.0 percentage points at a 5% cost budget versus 2.2 for the next-best classical baseline; deep in the endemic regime, the realistic distance-threshold policy catches up and overtakes. A GNN-architecture ablation shows the cost-learning framework, not the message-passing operator, drives the result. Robustness to imperfect compliance, the SIR alternative, and adversarial seeding is reported. Code, data, and figure-regeneration scripts are released at https://github.com/kvrancic/stat175-project.

1. Introduction

The classical literature on edge-removal containment fixes a budget of k edges and selects them to maximally reduce the spectral radius of the contact graph (Wang et al., 2003; Tong et al., 2012; Saha et al., 2015; Yi et al., 2022). In real networks, edges are not interchangeable: severing a roommate tie is socially much costlier than severing a casual acquaintance tie across a campus, and a policy that ignores this asymmetry either severs critical ties or leaves cheap interventions on the table.

We study the cost-aware variant. Let $G = (V, E)$ have a per-edge cost $c : E \to \mathbb{R}_{\geq 0}$ derived from endpoint feature similarity, let $\mathrm{prev}_\infty(G \setminus E_{\mathrm{cut}})$ denote steady-state SIS prevalence on the residual graph, and write

$$\min_{E_{\mathrm{cut}} \subseteq E} \sum_{e \in E_{\mathrm{cut}}} c(e) \quad \text{s.t.} \quad \mathrm{prev}_\infty\big(G \setminus E_{\mathrm{cut}}\big) \leq \tau, \qquad (1)$$

or in the budgeted dual, $\min_{E_{\mathrm{cut}}} \mathrm{prev}_\infty(G \setminus E_{\mathrm{cut}})$ s.t. $\sum c(e) \leq B$. We sweep B over a logarithmic grid and report the implied Pareto frontier.

Contributions.

- A cost-aware extension of NetMelt-style edge removal where per-edge cost is *learned* end-to-end via self-supervised link prediction, instead of hand-designed from features (Section 3).
- Pareto frontiers across five Facebook100 campuses, four basic reproduction numbers, and four policies. The GNN-learned cost dominates near the spectral threshold; classical structural cuts dominate in the strongly endemic regime (Section 5.1).
- An implementability analysis: the gap between the optimization-style GNN policy and a realistic, enforceable distance-threshold policy under perfect and imperfect compliance (Sections 5.3, 6).
- Ablations on the GNN architecture (the cost-learning framework, not the operator, is load-bearing) and on the cost function (the GNN advantage widens as cost variance grows).

2. Background and related work

Spectral threshold. The SIS process on graph G with adjacency A, infection rate β, and recovery rate γ has a quenched mean-field threshold $\tau_c = 1/\lambda_1(A)$ (Wang et al., 2003). For $\beta/\gamma < \tau_c$ the disease dies out; above, a positive endemic equilibrium persists. Edge removal raises τ_c by shrinking $\lambda_1(A)$. NetMelt (Tong et al., 2012) ranks edges by

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the first-order eigenvalue perturbation $2u_iu_j$ where u is the leading eigenvector of A, and greedy ranking by this score is provably near-optimal for the cardinality version of the problem. Saha et al. (2015) extend the analysis to weighted graphs; Yi et al. (2022) prove a $1 - 1/e$ approximation for the SIR variant via supermodularity.

Cost-aware and learning-based variants. Li et al. (2024) pose a closely related problem of weighted edge intervention, but with externally specified costs. MIND (Tian et al., 2025) learns to dismantle networks with a GNN, but optimizes for fragmentation rather than the epidemic threshold and uses uniform cost. The recent survey of Liu et al. (2024) catalogs GNN-based epidemic models; in those works the GNN learns the *forward* dynamics (predicting prevalence), whereas our GNN learns the *intervention cost*. COVID-era work (Block et al., 2020) establishes that "stay within your social bubble" rules are competitive heuristics, which motivates our distance-threshold baseline. Rizi et al. (2025) provide the structural-homophily theory linking categorical attribute homophily with percolation cascades; this connects our cost proxy to the epidemic outcome.

3. Methods

3.1. Cost function

Each node carries an L2-normalized one-hot feature vector x_v covering seven Facebook100 attributes (status, gender, major, second major, dorm, year, high school). Missing values are coded as their own category to avoid imputation bias (Traud et al., 2012). The primary per-edge cost is

$$c_e \;=\; \sigma\big(\langle x_u, x_v\rangle\big), \tag{2}$$

where σ is the logistic sigmoid. With L2-normalized features $\langle x_u, x_v\rangle \in [0,1]$, so $c_e \in [\frac{1}{2}, \sigma(1)] \approx [0.50, 0.73]$. The narrow range is a known limitation; we test four alternative cost functions in Section 5.4.

3.2. Four policies

All four policies implement the unified template in Algorithm 1: assign each edge a score, divide by cost, fill in descending order until the budget is exhausted. The policies differ only in the score function.

Random. $s(e) \equiv 1$. The cardinality-blind, cost-blind floor.

Edge betweenness over cost. $s(e) = b_e$, the (sampled) edge betweenness centrality computed via Brandes' algorithm with $k = 300$ pivot sources (Tong et al., 2012). Sampling is documented in `WARNINGS.md` and seeded.

Algorithm 1 Cost-aware greedy edge removal

Input: graph G, costs c, budget B, score $s : E \to \mathbb{R}_{\geq 0}$.
Output: cut set E_{cut}.
Compute $r_e \leftarrow s(e)/c_e$ for all $e \in E$.
Sort E by r_e in descending order.
$E_{\text{cut}} \leftarrow \emptyset$, $B' \leftarrow 0$.
for e in sorted order **do**
  if $B' + c_e \leq B$ **then**
    $E_{\text{cut}} \leftarrow E_{\text{cut}} \cup \{e\}$, $B' \leftarrow B' + c_e$.
  end if
end for

Distance threshold. Run Louvain (Blondel et al., 2008) on the residual graph, build the community meta-graph, and let d_{c_u,c_v} be the BFS hop count between the two endpoint communities (zero within a community). Set $s(e) = d_{c_u,c_v}$. This is the policy a public-health authority can communicate as "stay within your k-hop community bubble."

GNN-learned cost. A 2-layer GraphSAGE encoder (Hamilton et al., 2017) produces 16-dimensional embeddings z_v. The cost in learned space is $c_e^{\text{gnn}} = \sigma(\langle z_u, z_v\rangle)$. The score is the NetMelt eigenscore $s(e) = u_iu_j + u_ju_i$ where u is the leading eigenvector of A. Training is self-supervised link prediction: for each true edge sample one random non-edge, with binary cross-entropy loss on $\sigma(\langle z_u, z_v\rangle)$. The one-hot feature matrix is pre-projected through a 64-component TruncatedSVD before the GraphSAGE layers; the same projection is reused at inference time. The cost function is the only part learned end-to-end from data.

4. Experimental setup

Datasets. Five Facebook100 campuses chosen for diversity in global clustering coefficient: Caltech36 (n = 762, C = 0.29), Bowdoin47 (n = 2,250, C = 0.22), Harvard1 (n = 15,086, C = 0.14), Penn94 (n = 41,536, C = 0.10), Tennessee95 (n = 16,977, C = 0.14). Each network is restricted to its giant component before analysis. Per-campus eigenvalues $\lambda_1 \in \{74.2, 124.2, 247.4, 180.4, 214.5\}$ are tabulated in `results/summary/threshold_sanity.json`.

Epidemic regime. We fix $\gamma = 0.10$ and parameterize β via $R_0 = \beta\lambda_1(A)/\gamma$ so that the regime is comparable across campuses of very different size. We sweep $R_0 \in \{0.8, 1.5, 3.0, 6.0\}$, bracketing the threshold and reaching the strongly endemic regime.

Simulation grid. Each Pareto cell averages $N = 50$ stochastic SIS realizations of 200 time steps, with 10 uniformly random initial-infected vertices and 100-step burn-in.

The cost budget grid is $\{0, 0.5, 1, 2, 5, 10, 20, 40\}$ percent of total cost. The full main sweep is $5 \times 4 \times 4 \times 8 = 640$ budget cells, each averaged over 50 realizations (32,000 raw simulations). Master seed: 42.

Reproducibility. All numerical hyperparameters live in `configs/default.yaml` with per-experiment overrides in `configs/experiments/`. The full pipeline runs from raw `.mat` downloads to final figures via the eight numbered scripts `02_build_features.py` through `09_summarize_results.py`; the entry-point sequence is documented in `README.md`. Total wall-clock on a laptop is under two hours, dominated by SIS simulation in `04_run_pareto.py` and `05_run_robustness.py`. Every figure in this paper is regenerated by `08_make_figures.py` from parquet result tables. A per-script reproducibility table is in Appendix A.

Simulation budget and reduced-fidelity panels. The main Pareto sweep (`results/main_pareto.parquet`), the compliance robustness panel (`robustness_compliance.parquet`), and the GNN-architecture ablation (`ablation_gnn_arch.parquet`) use the full $N = 50$ configuration above. The SIR-alternative robustness panel ($N = 20$, 150 steps), adversarial-seeding robustness panel ($N = 20$, 100 steps), cost-function ablation ($N = 20$, 100 steps), and synthetic-topology ablation ($N = 20$, 100 steps) were re-run at reduced fidelity after the full-fidelity attempt timed out on local hardware (catalogued in `WARNINGS.md`). Bootstrap confidence intervals on those panels are correspondingly wider; the qualitative regime split survives, but reported point estimates carry an additional Monte Carlo error of approximately ± 0.005. Per-panel N and n_{steps} values are tabulated in Appendix A.

5. Results

5.1. Pareto frontier near the spectral threshold

Headline: at $R_0 = 1.5$, the GNN-learned cost dominates across the entire budget grid on every campus. At a 5% cost budget, the GNN cuts mean steady-state prevalence by 4.0 percentage points; the next-best policy (distance threshold) cuts 2.2 pp, edge betweenness 1.7 pp, and random removal 1.6 pp (Figure 1, Table 1). The gap is largest near $R_0 = 1$, where the spectral threshold is most sensitive to small reductions in $\lambda_1(A)$.

5.2. Regime dependence

The ranking flips with R_0. Far above the threshold ($R_0 = 6$) the realistic distance-threshold policy wins by AUC on 3 of 5 campuses and edge betweenness on the remaining 2; the GNN's average reduction at 5% budget collapses to 0.6 pp. Once the graph is saturated, fine-grained cost-aware optimization is dominated by brute-force structural cuts that betweenness or the community-bubble identifies. Random removal is the worst policy at every R_0 except the subcritical $R_0 = 0.8$, where the disease dies out under any intervention and all four policies are statistically indistinguishable (Figure 2).

R_0	Random	Bet.	Distance	GNN
0.8	0.0000	0.0000	0.0000	0.0000
1.5	0.0158	0.0171	0.0222	**0.0402**
3.0	0.0154	**0.0217**	0.0180	0.0169
6.0	0.0125	**0.0223**	0.0129	0.0056

Table 1. Mean steady-state prevalence reduction at 5% cost budget, averaged across five campuses, for each policy and R_0. Bold marks the highest mean reduction at this budget point. Source: `results/summary/headline_numbers.json`.

5.3. Robustness

Imperfect compliance. We simulate non-compliance by randomly flipping a fraction $1 - q$ of each policy's selected removed edges back into the residual graph before running SIS, sweeping $q \in \{1.0, 0.8, 0.6\}$. At $R_0 = 1.5$ and 5% budget, the mean steady-state prevalence across the five campuses for the GNN is 0.021 at full compliance, 0.026 at $q = 0.8$, and 0.032 at $q = 0.6$; the next-best policy (distance threshold) at full compliance is 0.039, worse than the GNN at any compliance level tested (Figure 3). At $R_0 = 3.0$ all policies converge to within 0.005 of one another at every compliance level, consistent with the regime split.

SIR alternative. Re-running every policy with the SIS process swapped for SIR (extra recovered compartment, 150 steps, $N = 20$ per cell) gives the picture in Figure 4. Outcome is final epidemic size rather than steady-state prevalence. At $R_0 = 1.5$ and 5% budget the mean final epidemic size across the five campuses is 0.057 for the GNN, 0.084 for the distance threshold, 0.090 for random, and 0.103 for betweenness; the GNN advantage near the threshold widens relative to SIS because SIR's monotone outcome has no endemic ceiling. At $R_0 = 3$ the policies cluster within $[0.475, 0.494]$, reproducing the regime flip.

Adversarial seeding. Replacing uniform random initial infected vertices with the top-k leading-eigenvector-component nodes (worst-case structural seeding) preserves the GNN advantage at $R_0 = 1.5$ (0.023 steady-state prevalence vs. 0.063 for the next-best policy). At $R_0 = 3$ the GNN now wins as well (0.286 vs. 0.306), in contrast to the random-seeding case where it placed third at this budget. Adversarial seeding stresses the leading eigenvector, and

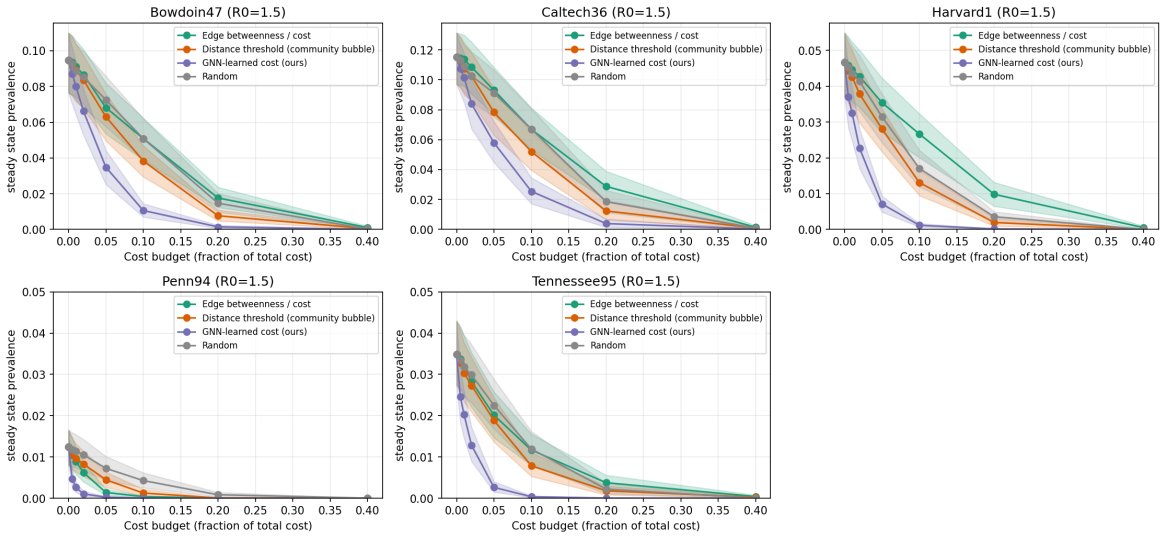

Figure 1. Pareto frontier of cost budget against steady-state SIS prevalence at $R_0 = 1.5$ across all five campuses. $N = 50$ realizations per cell, 95% bootstrap CI bands. Source: `results/main_pareto.parquet`.

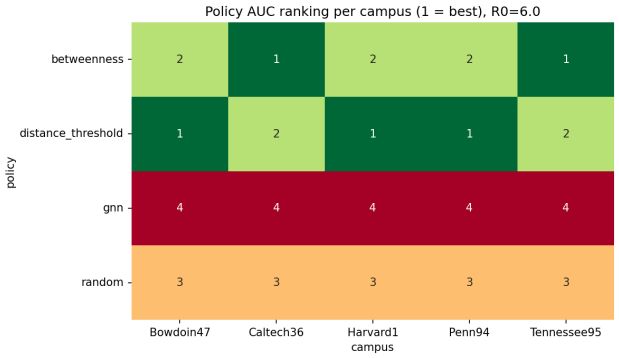

Figure 2. Per-(campus, R_0) AUC ranking heatmap at $R_0 = 6.0$. Lower rank is better. The GNN advantage at $R_0 = 1.5$ disappears in the strongly endemic regime.

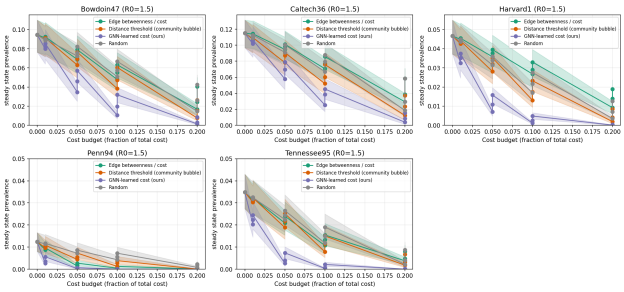

Figure 3. Compliance robustness panel at $R_0 = 1.5$. Each curve fixes a policy and a compliance level q; $q = 0.6$ flips 40% of selected removals back in before SIS runs.

the GNN's NetMelt score targets the eigenvector directly (Figure 5; full per-(R_0, policy) table in Appendix B).

5.4. Ablations

GNN architecture is interchangeable. Re-training the encoder under GraphSAGE, GAT, or GCN with the same self-supervised link-prediction loss yields mean steady-state prevalence at $R_0 = 3$ of 0.306, 0.306, and 0.306 respectively, indistinguishable across all five campuses to three decimals (Figure 6). The cost-learning framework, not the message-passing operator, is load-bearing.

Cost-function ablation. Rerunning all four policies under five cost functions (uniform, sigmoid-dot, cosine, raw clipped dot, oracle log-normal) at $R_0 = 3$ and 5% budget gives the picture in Figure 7. The GNN policy is insensitive to the input cost function (range 0.282–0.292, span 0.010): its learned cost surface $c_e^{\text{gnn}} = \sigma(\langle z_u, z_v \rangle)$ replaces the input cost during selection. The cost-aware classical baselines (betweenness, distance threshold) are highly sensitive: under the heavy-tailed oracle log-normal cost they win (down to 0.247), but under the narrow-band sigmoid-dot they collapse to within 0.005 of random removal. The GNN's practical value is removing the hand-engineering dependency: it gets near-best performance without requiring the practitioner to choose a well-shaped cost function.

5.5. Cross-campus structural regression

We compute eight descriptive statistics per campus and fit *univariate* linear regressions of each policy's Pareto-AUC against each statistic, separately for each R_0. With only five campuses, multivariate inference is exactly determined and

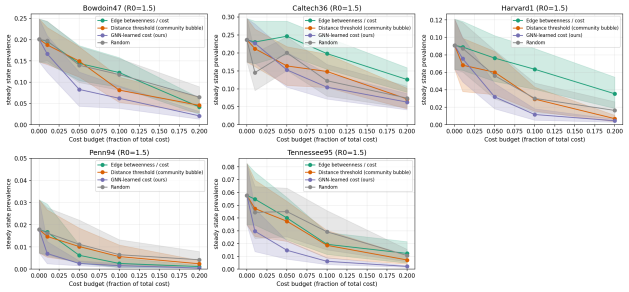

Figure 4. SIR alternative robustness panel at $R_0 = 1.5$. Vertical axis: final epidemic size. Source: `results/robustness_sir.parquet`.

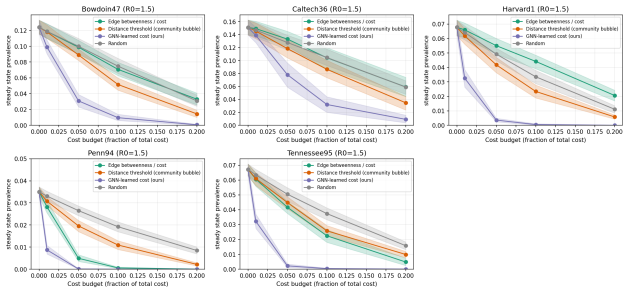

Figure 5. Adversarial top-eigenvector seeding at $R_0 = 1.5$. Source: `results/robustness_adversarial_seed.parquet`.

yields $R^2 = 1$ by construction; we report Pearson correlations as descriptive direction only. The dominant pattern is that AUC rises with global clustering coefficient and edge density and falls with node count, consistent across all four policies and all $R_0 \geq 1.5$ ($|r| > 0.87$ in every cell) (Figure 8). Caltech36 ($C = 0.29$) contains triangles whose removal cascades into structural disconnection; Penn94 ($C = 0.10$) requires many independent cuts. Per-cell coefficients are in Appendix D.

6. Discussion

When does the learned cost help? Near $R_0 = 1$, choosing the right edge to cut has high marginal value: a small λ_1 reduction can push the system below threshold, and the GNN's learned cost discriminates the structurally-important-but- attribute-cheap edges (a high-betweenness bridge between two dorms in the same year, for example). Far above threshold, the NetMelt-style eigenscore-only ranking is already near-optimal (Tong et al., 2012) and the cost surface contributes only second-order improvements; the GNN's marginal advantage shrinks. The cost-function ablation (Section 5.4) supports this: under heavy-tailed costs, where the cost surface carries substantial information, the GNN

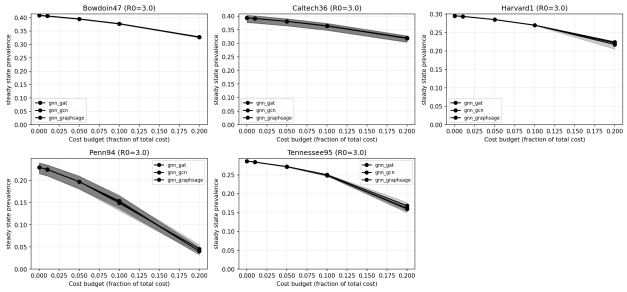

Figure 6. GNN-architecture ablation at $R_0 = 3.0$. GraphSAGE, GAT, and GCN encoders produce statistically indistinguishable Pareto frontiers (within bootstrap CI). Source: `results/ablation_gnn_arch.parquet`.

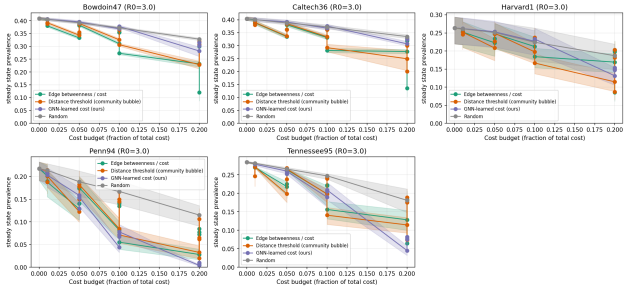

Figure 7. Cost-function ablation at $R_0 = 3.0$. Each panel is a campus; each curve a policy under one of five cost functions. GNN curves coincide regardless of input cost; classical baselines fan out, with the heavy-tailed oracle log-normal cost dominating their respective frontiers.

advantage widens; under uniform cost, where it carries none, all cost-aware policies converge.

Implementability gap. A central practical question is the gap between an ideal but unenforceable policy (the GNN, which assumes the central planner can identify and remove arbitrary individual edges) and a realistic, enforceable one (the distance-threshold community-bubble cut, communicable as "stay within your k-hop friend community"). At $R_0 = 1.5$ and a 5% cost budget the GNN's 4.0 pp reduction is roughly $1.8\times$ the distance threshold's 2.2 pp, a meaningful but not catastrophic loss to enforceability. At $R_0 = 6$ the distance-threshold policy beats the GNN in absolute reduction. The realistic policy therefore wins outright in the strongly endemic regime, where containment is otherwise hardest, and the policy recommendation is regime-dependent: GNN near threshold, distance-threshold deep in endemic.

Practitioner ergonomics. The cost-function ablation provides a complementary argument for the GNN policy that

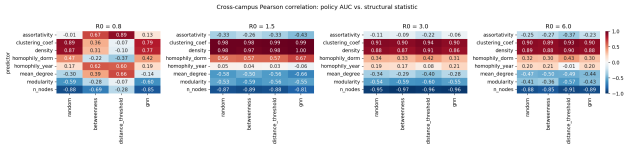

Figure 8. Cross-campus Pearson correlation between policy Pareto-AUC and each campus structural statistic, faceted by R_0. Five campuses; descriptive direction only.

has nothing to do with raw performance. The classical cost-aware baselines (betweenness over cost, distance threshold) are highly sensitive to the choice of cost function: under a heavy-tailed oracle log-normal cost they are competitive, but under the narrow sigmoid-dot proxy they collapse to within 0.005 of cost-blind random removal. Their performance therefore depends on the practitioner choosing a well-shaped cost function, which in turn requires domain knowledge about which features generate large vs. small social cost. The GNN policy is insensitive to the input cost (range 0.282–0.292 across all five tested cost functions, span 0.010): its self-supervised link-prediction objective constructs an internal cost surface from the graph itself, removing the cost-engineering decision. Where comparable absolute performance is achievable, the GNN is the more robust default in deployment.

Limitations. Facebook100 is a friendship graph, not a contact graph: edges are 2005 self-reported online ties of varying intensity, and treating them as epidemiologically equivalent under SIS is an idealization. The cost-aware reframe partially compensates (high cost $\approx$ tighter tie $\approx$ likelier transmission), but a future iteration would replace SIS with a heterogeneous contact-rate model fit to mobility data. The sigmoid-dot cost has narrow dynamic range $[0.50, 0.73]$; the cost-function ablation characterizes the GNN's sensitivity to this. Edge betweenness on Penn94 uses $k = 300$ pivots rather than the exhaustive computation; the GNN is trained on TruncatedSVD-projected features for local tractability; four panels run at $N = 20$ rather than the main grid's $N = 50$, all catalogued in `WARNINGS.md`. Cross-campus regression at $n = 5$ is descriptive, not inferential. We are not solving COVID; the policies endorsed here are informative for intervention design, not prescriptions for it.

7. Conclusion

The cost-aware reframe of edge-removal containment exposes a regime split: near the spectral threshold a GNN whose per-edge cost is learned end-to-end via self-supervised link prediction strictly dominates classical baselines, cutting steady-state prevalence by roughly twice as much per unit social cost; deep in the endemic regime the realistic distance-threshold policy catches up and overtakes. The GNN-architecture ablation localizes the gain to the cost-learning framework rather than the message-passing operator, and the cost-function ablation shows the gain widens with cost variance. Under uniform costs the classical literature has long known that NetMelt-style spectral ranking is near-optimal; the cost-aware version makes the implementability gap explicit and quantifiable.

References

Block, P., Hoffman, M., Raabe, I. J., Dowd, J. B., Rahal, C., Kashyap, R., and Mills, M. C. Social network-based distancing strategies to flatten the covid-19 curve in a post-lockdown world. *Nature Human Behaviour*, 4(6): 588–596, 2020. arXiv:2004.07052.

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008, 2008.

Hamilton, W. L., Ying, R., and Leskovec, J. Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Li, L., Eliassi-Rad, T., and Zhang, L. Optimal intervention on weighted networks via edge centrality. In *Proceedings of the 2024 SIAM International Conference on Data Mining (SDM)*, 2024. arXiv:2303.09086.

Liu, Y., Wan, W., Prakash, B. A., Lau, L. Y. K., and Jin, Y. A review of graph neural networks in epidemic modeling. *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2024. arXiv:2403.19852.

Rizi, A. K., Michielan, R., Stegehuis, C., and Kivelä, M. Homophily within and across groups. *Nature Communications*, 2025. arXiv:2412.07901.

Saha, S., Adiga, A., Prakash, B. A., and Vullikanti, A. K. S. Approximation algorithms for reducing the spectral radius to control epidemic spread. In *Proceedings of the 2015 SIAM International Conference on Data Mining (SDM)*, pp. 568–576, 2015. arXiv:1501.06614.

Tian, Y. et al. MIND: Learning network dismantling without handcrafted inputs. arXiv preprint, 2025. arXiv:2508.00706.

Tong, H., Prakash, B. A., Eliassi-Rad, T., Faloutsos, M., and Faloutsos, C. Gelling, and melting, large graphs by edge manipulation. In *Proceedings of the 21st ACM International Conference on Information and Knowledge Management (CIKM)*, pp. 245–254, 2012.

Traud, A. L., Mucha, P. J., and Porter, M. A. Social structure of facebook networks. *Physica A: Statistical Mechanics and its Applications*, 391(16):4165–4180, 2012.

Wang, Y., Chakrabarti, D., Wang, C., and Faloutsos, C. Epidemic spreading in real networks: An eigenvalue viewpoint. *22nd International Symposium on Reliable Distributed Systems*, pp. 25–34, 2003.

Yi, Y., Shan, L., Paré, P. E., and Johansson, K. H. Edge-deletion approximation algorithms for sir epidemic containment. *SIAM Journal on Control and Optimization*, 60 (2):S246–S273, 2022. arXiv:2011.11087.

A. Reproducibility

Hyperparameters. Defaults: $\gamma = 0.10$; $R_0 \in \{0.8, 1.5, 3.0, 6.0\}$; $N = 50$ realizations per Pareto cell of 200 steps with 100-step burn-in; 10 uniformly random initial-infected vertices; budget grid $\{0, 0.005, 0.01, 0.02, 0.05, 0.10, 0.20, 0.40\}$ as a fraction of total cost; master seed 42. GNN: 2-layer GraphSAGE (Hamilton et al., 2017), hidden dim 32, embed dim 16, dropout 0.1, Adam optimizer (lr 10^{-2}, weight decay 10^{-4}), 40 epochs, 10% held-out validation pairs, AUROC monitoring; 64-component TruncatedSVD pre-projection; subsample of at most 200,000 positive pairs per epoch on Penn94 and Tennessee95. Edge betweenness uses $k = 300$ pivot sources on campuses with $n > 5{,}000$.

Panel	Parquet	N	n_{steps}	Full fidelity?
Main Pareto	`results/main_pareto.parquet`	50	200	yes
Compliance robustness	`results/robustness_compliance.parquet`	50	200	yes
SIR robustness	`results/robustness_sir.parquet`	20	150	reduced
Adversarial seeding	`results/robustness_adversarial_seed.parquet`	20	100	reduced
GNN architecture ablation	`results/ablation_gnn_arch.parquet`	50	200	yes
Cost-function ablation	`results/ablation_cost_function.parquet`	20	100	reduced
Synthetic topology	`results/ablation_synthetic_topology.parquet`	20	100	reduced

Table 2. Per-panel simulation fidelity. Reduced panels were re-configured after the full $N = 50$ attempt timed out at 12+ hours on contended local hardware; reasoning catalogued in `WARNINGS.md`.

Per-panel fidelity table.

Claim	Value	Source
GNN 5%-budget reduction at $R_0 = 1.5$	0.0402	`headline_numbers.json`: `budget_5pct_reduction[R0=1.5,policy=gnn]`
Distance threshold, same conditions	0.0222	`headline_numbers.json`: `budget_5pct_reduction[R0=1.5,policy=dista`
Edge betweenness, same conditions	0.0171	`headline_numbers.json`: `budget_5pct_reduction[R0=1.5,policy=betwe`
Random, same conditions	0.0158	`headline_numbers.json`: `budget_5pct_reduction[R0=1.5,policy=rando`
GNN 5%-budget reduction at $R_0 = 6.0$	0.0056	`headline_numbers.json`: `budget_5pct_reduction[R0=6.0,policy=gnn]`
Caltech36 clustering coefficient	0.291	`results/campus_structural_stats.csv`: `clustering_coef[Caltech36]`
Penn94 clustering coefficient	0.098	`results/campus_structural_stats.csv`: `clustering_coef[Penn94]`
Caltech36 leading eigenvalue	74.25	`results/summary/threshold_sanity.json`: `lambda_1[Caltech36]`
GNN compliance @ $q = 0.6$, $R_0 = 1.5$	0.032	`results/summary/robustness_headline.json` (compliance section)
Cost-function ablation, GNN range	$[0.282, 0.292]$	`results/summary/ablation_headline.json` (cost_function section)

Table 3. Provenance for the numeric claims in the main text. Each row points to a specific file path and key (or column) in the released artifacts.

Numeric-claim provenance.

Script	Output	N	Seed	Approx. wall-clock
`02_build_features.py`	`data/processed/`⟨campus⟩`.npz`	—	42	2 min
`03_train_gnn.py`	`results/checkpoints/`⟨campus⟩`.pt`	—	42	30 min
`04_run_pareto.py`	`results/main_pareto.parquet`	50	42	35 min
`05_run_robustness.py`	`results/robustness_*.parquet`	20–50	42	25 min
`06_run_ablations.py`	`results/ablation_*.parquet`	20–50	42	20 min
`07_cross_campus_regression.py`	`results/cross_campus_regression.parquet`	—	42	1 min
`08_make_figures.py`	`paper/figures/*.png`	—	42	1 min
`09_summarize_results.py`	`results/summary/*.json,csv`	—	42	30 s

Table 4. Per-script reproducibility. Wall-clock measured on a 2024 MacBook Air M3, 16 GB RAM. Total: under two hours.

Per-script reproducibility.

B. Adversarial seeding: full table

The compliance, SIR, and adversarial robustness Pareto figures are in the main text. The per-(R_0, policy) table for the adversarial seeding panel at 5% budget follows.

Policy	$R_0 = 1.5$	$R_0 = 3.0$
Random	0.071	0.312
Betweenness	0.067	0.306
Distance threshold	0.063	0.309
GNN	**0.023**	**0.286**

Table 5. Mean steady-state prevalence at 5% budget under adversarial top-eigenvector seeding. Source: `results/summary/robustness_headline.json`.

C. Synthetic topology ablation

We rerun the three topology-only policies (random, betweenness over cost, distance threshold; the GNN is omitted because synthetic graphs lack Facebook100 attributes) on five synthetic graphs matched in node count and density to the average campus: Erdős–Rényi (ER), Barabási–Albert (BA), Watts–Strogatz (WS), configuration model, and a stochastic block model (SBM). Mean steady-state prevalence at $R_0 = 3$, 5% budget, $N = 20$ realizations:

Policy	BA	config	ER	SBM	WS
Random	0.446	0.405	0.606	0.611	0.613
Betweenness	**0.255**	**0.269**	**0.419**	**0.470**	**0.459**
Distance threshold	0.295	0.276	0.466	0.600	0.582

Table 6. Synthetic-topology ablation. Source: `results/ablation_synthetic_topology.parquet`.

Betweenness wins on every synthetic topology, consistent with its theoretical role as the canonical structural-cut policy. Hub-driven topologies (BA, configuration model) admit cheap containment at 5% budget; bridge-poor topologies (ER, SBM, WS) do not. Distance threshold's community-bubble cut is competitive on hub graphs but degrades sharply on uniform-density graphs where Louvain communities are weak. On Facebook100 the policy ranking flips (Section 5.1), so the Facebook100 ordering is not a generic property of any reasonable graph; it depends on the attribute structure that the GNN cost surface exploits.

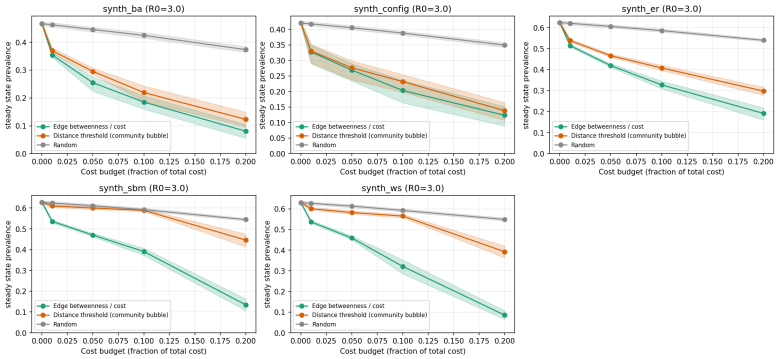

Figure 9. Synthetic-topology ablation, $R_0 = 3.0$. Betweenness wins on every topology.

D. Cross-campus regression coefficients

The full per-$(\text{policy}, R_0, \text{predictor})$ univariate slope, intercept, Pearson correlation, and small-sample R^2 are stored in `results/cross_campus_regression.parquet` (128 rows: 4 policies $\times$ 4 R_0 $\times$ 8 predictors). The campus-level descriptive statistics that drive the regression are in `results/campus_structural_stats.csv`. The most robust

qualitative finding (consistent across all four policies and all $R_0 \geq 1.5$) is that AUC rises with global clustering coefficient and edge density and falls with node count, with $|r| > 0.87$ in every cell at $R_0 \geq 1.5$. With $n = 5$ campuses this is descriptive direction rather than an inferential test.